

MATH 212

HOMEWORK DUE WEEK 6 MONDAY

Winter 2026

Prof. Chowdhury

■ Question 6001.

Q6

Prove that the polynomials $\{q_n(x)\}$ where $q_n(x) = \prod_{j=0}^{n-1} (x - x_j)$ are linearly independent.

HINT: use induction.

■ Question 6002.

□

Let k be a positive integer. Suppose that $p \in \mathcal{P}_{2k+1}$ such that

$$p(-1) = p(1) = 0$$

and

$$p^{(2i-1)}(\pm 1) = 0 \quad \text{for } i = 1, \dots, k$$

Prove that $p \equiv 0$.

■ Question 6003.

Q15

Note: This question is optional. You do not need to submit an answer.

Suppose that p_n is the polynomial of degree $\leq n$ that interpolates $f(x)$ at the distinct points $x_0, x_1, \dots, x_n$. Develop a conjecture about what p_n converges towards as the interpolating points $x_1, \dots, x_n$ all approach x_0 simultaneously, and explain your thinking.

HINT: use the definition of $E_n(x)$ and the Interpolation Error formula (Cauchy) we developed in class.

■ Question 6004.

Q21

Suppose S_3 is the natural cubic spline interpolant to $\{(x_j, f_j)\}_{j=0}^n$, and σ is any $C^2[x_0, x_n]$ function that also interpolates the same data. Then show that

$$\|S_3''\|_2 \leq \|\sigma''\|_2.$$

■ Question 6005.

□

Interpolation connects data pairs using polynomial functions, where the degree of the polynomials is chosen so that all pairs are matched exactly. However, real-world data tend to be inaccurate, in which case exact matching is not necessarily desirable. Instead, we might know (or suspect) that a general type of function

(not necessarily polynomial) underlies the data, and we want to choose a particular function of this type to represent the data. Identifying a particular “model” function for the data would allow us not only to predict the behavior of the system at points not represented in the collected dataset, but to study the system more generally. This process of identifying a particular model function for a set of data is called **data fitting** (or “regression”), and we usually choose the model function that “best” fits the data in some sense.

Over the first four weeks, you have seen various linear regression techniques. Recall that the vector $\vec{x}^*$ that minimizes the “Least Square Error” (LSE) $\|A\vec{x} - \vec{b}\|$ is found by solving the Normal equation (you do not need to prove this)

$$A^T A \vec{x}^* = A^T \vec{b}.$$

This exercise will ask you to prove some of the related linear algebra results.

Suppose A is an $m \times n$ matrix (not necessarily square). Prove the following linear algebra results:

(a) $(\text{Col}(A))^\perp = \text{Nul}(A^T).$

Note: $\text{Nul}(A)$ is the nullspace or kernel of A which is defined as $\text{Nul}(A) = \{\vec{v} \in \mathbb{R}^n \mid A\vec{v} = \vec{0}\}.$

$\text{Col}(A)$ is the column space or range of A which is defined as $\text{Col}(A) = \{A\vec{v} \mid \vec{v} \in \mathbb{R}^n\}.$

$\text{rank}(A)$ is the dimension of $\text{Col}(A).$

(b) $\text{Nul}(A) = \text{Nul}(A^T A).$

HINT: It is easy to show that $\text{Nul}(A) \subseteq \text{Nul}(A^T A)$. To show the opposite containment $\text{Nul}(A^T A) \subseteq \text{Nul}(A)$, use orthogonal complements.

(c) $\text{rank}(A) = \text{rank}(A^T A).$

Hence if A has full rank, we get $\text{Nul}(A^T A) = \{0\}$. Consequently, $A^T A$ is invertible, and we have an exact solution to the normal equation $\vec{x}^* = (A^T A)^{-1} A^T \vec{b}.$